

SANCHEZ

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SUMMARY

Computer Science graduate from UTRGV with hands-on experience in systems programming, full-stack web development, machine learning, and low-level programming. Delivered independent projects including a deployed interactive portfolio and a polyp detection model achieving mAP50 of 0.86. Actively developing Java and Spring Boot proficiency with cloud exposure via Microsoft Azure. Bilingual in English and Spanish.

TECHNICAL SKILLS

Languages: Java, Python, C++, C, C#, Ruby, Rust, Assembly x86, JavaScript

Web / Frontend: React, HTML, CSS, Flutter, Phaser3

Tools: Git, GitHub, Jira, Vim, Android Studio, Unity, Wireshark, VirtualBox, Figma

Cloud & OS: Microsoft Azure, UNIX/Linux, PowerShell, CMD

PROJECTS

Personal Portfolio Website 2026

React | Phaser 3 | JavaScript | Vim

- Built an interactive browser-based portfolio as a 2D arcade game, replacing a static resume with an explorable experience, by developing it entirely in React and Phaser 3 through self-directed learning outside of formal coursework. Deployed to production via Vercel with features including a day/night cycle, dynamic rain, a token-based gift shop, and a phone booth contact panel.

2D Top-Down Game In Progress

Unity | C# | AI Algorithms

- Currently developing a 2D top-down game in Unity using C# with AI-driven enemy behavior and custom level design. Implementing state machine logic and core gameplay systems including collision detection, player movement, and scene management.

Efficient Polyp Detection 2025

Python | YOLOv12 | Weights & Biases

- Achieved a mAP50 of 0.86 on the Kvasir-SEG dataset for colorectal polyp detection by training and tuning a YOLOv12m object detection model with image augmentation, preprocessing, and hyperparameter optimization. Tracked all experiments using Weights & Biases and presented findings as a senior capstone project.

Multiplayer Game Server & Client 2024

Python | Sockets | Multi-threading

- Enabled stable real-time gameplay for two concurrent players across three games by building a TCP socket server in Python with a three-way handshake, multi-threaded client handling, and structured logging, using no third-party networking libraries.

Unix Shell Implementation 2024

C | UNIX/Linux | Process Management

- Expanded shell functionality by extending a Unix shell in C with PATH resolution, dynamic prompts, persistent command history, subshell nesting, and child process management via fork and waitpid as part of an OS concepts course.

User-Mode Thread Scheduler 2024

C | Concurrency | OS Concepts

- Demonstrated OS-level concurrency understanding by implementing cooperative FIFO and Round-Robin schedulers entirely in user space using C, with semaphore-based synchronization and mutual exclusion built from scratch without kernel threading support, as part of an OS concepts course.

Ping Pong 2024

Assembly x86 | Low-Level Programming

- Implemented a two-player Ping Pong game in x86 Assembly by following low-level programming instruction, applying concepts including CPU register manipulation, memory addressing, collision detection, and real-time rendering with no high-level abstractions.

Grades Management System 2022

Ruby on Rails | Full-Stack | Authentication

- Built a full-stack web application in Ruby on Rails as a course project, implementing CRUD operations, secure user authentication, and role-based access control distinguishing between student and instructor views.

EDUCATION

University of Texas Rio Grande Valley MAY 2025

Bachelor of Computer Science | GPA: 2.81 | Minor: Asian Studies

South Texas College MAY 2016

Associate of Arts, Interdisciplinary Studies | GPA: 3.32

AWARDS & CERTIFICATIONS

- IBM Z 101 Certificate – Issued by IBM – Credential ID: 29fe6c0e-06d4-45ec-9221-1ba568424f77
- UTRGV Engaged Scholar Symposium - Honorary Mention & Distinguished Scholar Award (2018) for performance: "Tae Kwon Do - Korean Martial Arts"